

Databases and SQL

SQL 1 -- Introduction to DBMS & Keys

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About the Instructor

- **Rehber** -- Means "The Person who shows the right path" (fitting for a teacher!).
 - Working at **BlackRock** as a Low-Latency Platforms Engineer for Trading Execution.
 - Teaching for **3.5+ Years**. Mentored and coached **300+ students & learners**.
 - Likes to do Powerlifting & Endurance Work.
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What to Expect from This Module?

- Much easier than Data Structures & Algorithms.
- Get a fundamental grasp on Systems before LLD.

What to Expect from Rehber?

- Should not do any kind of formality. Give 100%.
- You are spending a lot of time and money. You need to get the most out of this.

What Rehber Expects from You

- Be regular and attentive in the classes.
- Complete the Assignments.
- Get all your doubts cleared.
- No Formality. It is a 2-way street.

How Can I Ace This Module?

- Attend the lectures.
 - Increase the PSP (Target **95%+**).
 - Achievable PSP as the Module & Assignments are Super Easy.
 - Easy segue into Low-Level Design.
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Software / Tools Needed (Pre-Requisite)

Homework for Today -- Install the following before the next class. See the [Appendix: Installation Guide](#) at the end of this document for detailed step-by-step instructions for macOS, Windows, and Linux.

We are standardizing everyone on **MySQL 8.4 LTS + MySQL Workbench 8.0.x + Sakila**. This gives the class a consistent environment and avoids the MySQL 9.x "Innovation" releases.

Component	Recommended	Purpose
MySQL Community Server	8.4 LTS	Actual database server
MySQL Workbench	8.0.47	GUI for SQL, schemas, ER diagrams, admin

Sakila Database	Current official download	Practice database
MySQL Shell	Optional	Command-line / advanced work

Note: The sample database is called **Sakila** (not "Sakela"). It is an official MySQL sample database designed for teaching and includes tables, views, stored procedures, functions, and triggers.

Module Roadmap

1. Keys
2. CRUD
3. Joins
4. Aggregate Queries
5. Subqueries
6. Indexing
7. Transactions
8. Window Functions
9. Schema Design

Let's Begin!

Data is Everywhere

What is the biggest thing that is happening in Tech right now? **Data is the currency.**

How Have We Been Storing Data Till Now?

Traditionally, data has been stored using **file systems** (e.g., CSV, TXT, JSON files).

Example: Getting the Average PSP of Learners

Given File: `Scaler.csv`

ID	Name	PSP
123	Rahul	93
456	Vikram	95
789	Vanshika	90

Pseudocode for Getting Average PSP:

1. Open the file
2. Read each line
3. Parse/split each line to extract the PSP column
4. Sum all PSP values
5. Divide by the number of records
6. Return the average

Issues with File Systems

Consider a larger dataset:

ID	Name	PSP	Attendance
123	Rahul	93	72
456	Vikram	95	90
789	Vanshika	90	77
321	Komal	72	94
654	Devinder	88	90

1. Inefficiency

- Searching, filtering, and aggregating data in flat files requires reading the entire file every time.
- No indexing or optimization is available.

2. Data Integrity Issues

- No enforcement of data types, constraints, or relationships.
- Duplicate or inconsistent data can easily creep in.

3. Concurrency Issues

- Multiple users or processes reading/writing the same file can lead to conflicts and data corruption.

4. Security Issues

- Difficult to control who can read or modify specific parts of the data.
- No fine-grained access control.

To solve these issues, we need to study **Databases & Database Management Systems (DBMS)**.

What is a Database?

A **Database** is a collection of related data -- e.g., music, books, blogs, user records, etc.

What is a Database Management System (DBMS)?

A **DBMS** is software that allows efficient management of a database. It:

- Helps us perform operations on data -- **CRUD** (Create, Read, Update, Delete).
 - Enforces the 4 rules we discussed above (Efficiency, Data Integrity, Concurrency, Security).
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Types of Databases

1. **Relational Databases** -- Data is stored in structured tables with defined relationships (e.g., MySQL, PostgreSQL, Oracle).
 2. **Non-Relational Databases** -- Data is stored in flexible formats like documents, key-value pairs, graphs, etc. (e.g., MongoDB, Redis, Neo4j).
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Properties of RDBMS (Relational Database Management System)

1. Data is Stored in the Form of Tables & Tables Can Be Related to Each Other

Persons Table:

ID	Name	Age
1	Max	28
2	Charles	27
3	Alex	25
4	Lance	28

SSN Table:

SSN	Issue Date
UAX123	21-Mar-03
VZ4560	4-Apr-12
MIQ234	30-Jan-09
COL023	18-Feb-18

Question: Can you tell which SSN belongs to Alex?

Without a relationship between the two tables, there's no way to answer this. This is why we need **Keys** to link tables together.

2. Every Row MUST Be Unique

No two rows in a table should be identical. Each row represents a distinct record.

3. All Values in a Column MUST Have the Same Data Type

ID	Name	Age
1	Max	28
2	Charles	27
3	Alex	25
4	Lance	28

Question: What is the Data Type of the `Age` column here? **Integer.**

4. Values Should Be ATOMIC

Each cell in a table should hold a single, indivisible value.

ID	Name	Mobile Number
----	------	---------------

1	Max	98300XXX
2	Charles	98670XXX

Question: How would you represent Max having 2 or more phone numbers?

You should **not** store multiple values in one cell (e.g., 98300XXX, 99100XXX). Instead, create a separate row or a separate table for phone numbers.

5. Column Sequence is Not Guaranteed

The order in which columns appear is not guaranteed by the RDBMS.

6. Row Sequence is Not Guaranteed

The order in which rows are returned is not guaranteed unless explicitly sorted.

7. Name of Every Column is Unique

No two columns in the same table can share the same name.

Keys

Data is in a table, but there's no way to use it effectively without **Keys**.

Keys help to uniquely identify a record in a table (or entity).

Types of Keys

1. Super Keys

Definition: Any combination of attributes (columns) that can uniquely identify a record.

Name	PSP	Email	Batch	Phone
Max	94	max@redbull.com	100	98300XXX
Charles	88	charles@ferrari.com	75	98670XXX
Alex	87	alex@williams.com	100	98670XXX

Examples of Super Keys:

- {Email}
- {Phone} (assuming unique in this dataset -- but be careful, Alex and Charles share the same phone here!)
- {Name, Batch}
- {Email, PSP}
- {Name, PSP, Email, Batch, Phone} (the entire row)
- ... and many more combinations.

A Super Key can have **redundant** attributes. As long as the combination is unique, it qualifies.

2. Candidate Keys

Definition: The **minimal** set of attributes required to uniquely identify a record. A Candidate Key is a Super Key with no redundant attributes.

Name	PSP	Email	Batch	Phone
Max	94	max@redbull.com	100	98300XXX
Charles	88	charles@ferrari.com	75	98670XXX
Alex	87	alex@williams.com	100	98670XXX

Examples of Candidate Keys: {Email} , {Name} (if unique)

Another Example:

Class Roll No	Batch Name	Email	PSP
1	SQL	max@redbull.com	83
1	LLD	charles@ferrari.com	84
2	HLD	alex@williams.com	82
2	SQL	lance@aston.com	81

Here:

- {Class Roll No} alone is **NOT** a Candidate Key (Roll No 1 appears twice).
- {Batch Name} alone is **NOT** a Candidate Key (SQL appears twice).
- {Class Roll No, Batch Name} together **IS** a Candidate Key (every combination is unique).
- {Email} is also a Candidate Key (every email is unique).

3. Composite Keys

Definition: Any key that consists of **more than one attribute** (size > 1).

Name	PSP	Email	Batch	Phone
Max	94	max@redbull.com	100	98300XXX
Charles	88	charles@ferrari.com	75	98670XXX
Alex	87	alex@williams.com	100	98670XXX

Examples of Composite Keys:

- {Name, Batch}
- {Email, PSP}
- {Class Roll No, Batch Name} (from the previous example)

A Composite Key can be a Super Key or a Candidate Key -- the defining feature is that it has **more than one column**.

4. Primary Key

Definition: The Candidate Key that is **chosen** by the database designer to uniquely identify records in a table.

- There can be **only one** Primary Key per table.
- It **cannot be NULL**.
- It **must be unique** for every row.
- It is used as the main reference for relationships with other tables.

***How to choose?** Pick the Candidate Key that is most stable, concise, and unlikely to change (e.g., an auto-incrementing `ID` or `Email`).*

5. Foreign Key

Definition: An attribute in one table that **references the Primary Key** of another table, creating a relationship between the two tables.

- Enforces **referential integrity** -- you cannot insert a value in the Foreign Key column that doesn't exist in the referenced Primary Key column.
- A table can have **multiple** Foreign Keys.

This is how we solve the earlier problem of linking the Persons table with the SSN table!

Summary

Concept	Description
Database	A collection of related data.
DBMS	Software to efficiently manage a database (CRUD, integrity, concurrency, security).
RDBMS	A DBMS that stores data in related tables with defined properties.
Super Key	Any combination of columns that uniquely identifies a row.
Candidate Key	A minimal Super Key (no redundant attributes).
Composite Key	A key made up of more than one column.
Primary Key	The chosen Candidate Key for the table (unique, not null, one per table).
Foreign Key	A column referencing the Primary Key of another table.

Homework

1. Install **MySQL Community Server 8.4 LTS**.
2. Install **MySQL Workbench 8.0.47**.
3. Download and set up the **Sakila Database**.
4. Record your MySQL root password somewhere safe.
5. Run the verification checks listed in the Installation Guide below.

See the full installation walkthrough in the Appendix below.

Appendix: Installation Guide

Official Download Links

1. **MySQL Server 8.4 LTS** -- [Download MySQL Community Server](#)
 2. **MySQL Workbench** -- [Download MySQL Workbench](#)
 3. **Sakila Database** -- [Download Sakila from MySQL's Example Databases page](#)
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macOS Setup

Step 0: Determine Your Mac Architecture

Go to: **Apple menu -> About This Mac**

You will see either:

- **Chip: Apple M1/M2/M3/M4...** -> Apple Silicon / ARM
- **Processor: Intel...** -> Intel / x86_64

For newer Macs, choose **ARM64**.

Step 1: Download & Install MySQL Server 8.4 LTS

1. Go to: [Download MySQL Community Server 8.4 LTS](#)
2. Choose: **macOS -> DMG Archive**
 - Apple Silicon Mac -> `macos15-arm64.dmg`
 - Intel Mac -> `macos15-x86_64.dmg`

The macOS 15 packages are compatible with macOS 14 Sonoma as well.

3. Open the downloaded `.dmg`.
4. Open the MySQL `.pkg`.
5. Continue through the installer.
6. Accept the license.
7. Choose the default installation.
8. **Set a root password** and save it somewhere safe.

```
Root username: root
Root password: <your chosen password>
```

Step 2: Verify MySQL Server is Running

Open **System Settings -> MySQL** to check if the server is running. You can start/stop it from there.

Test from Terminal:

```
mysql --version
# Expected: mysql Ver 8.4.x for macos...

mysql -u root -p
# Enter your root password
```

Once connected, run:

```
SELECT VERSION();  
SHOW DATABASES;
```

Exit with:

```
exit;
```

Step 3: Download & Install MySQL Workbench

1. Go to: [Download MySQL Workbench](#)
2. Choose the appropriate package:
 - o Apple Silicon -> **macOS (ARM, 64-bit), DMG Archive**
 - o Intel Mac -> **macOS (x86, 64-bit), DMG Archive**
3. Download the `.dmg` .
4. Double-click it.
5. Drag **MySQL Workbench** into **Applications**.
6. Open from: **Applications -> MySQL Workbench**.

Step 4: Connect Workbench to MySQL Server

1. Open MySQL Workbench.
2. On the home screen, click the **+** next to **MySQL Connections**.
3. Create a connection:

```
Connection Name: Local MySQL 8.4  
Hostname:        127.0.0.1  
Port:            3306  
Username:        root
```

4. Click **Test Connection** and enter your root password.
5. You should see: "**Successfully made the MySQL connection**".
6. Click **OK**.

Workbench 8.0 vs Server 8.4 Note: MySQL Workbench 8.0 is developed/tested with MySQL Server 8.0. It can connect to Server 8.4 and later, but some Workbench features may not function with later server versions. For our course (queries, tables, joins, constraints, views, procedures, etc.), this setup works well.

Step 5: Download the Sakila Database

Sakila is especially good for a DBMS/SQL course. It contains realistic tables such as:

```
actor , address , category , city , country , customer , film , film_actor ,  
film_category , inventory , language , payment , rental , staff , store
```

It also includes views, stored procedures, stored functions, and triggers.

1. Go to: [Sakila Database -- Official MySQL Download Page](#)
2. Under **Example Databases** -> **sakila database**, download the **Zip** (simpler for most users; TGZ also works).
3. Extract the archive. You will get:

```
sakila-db/  
  sakila-schema.sql  
  sakila-data.sql  
  sakila.mwb
```

- `sakila-schema.sql` -- Creates the database schema (tables, views, etc.).
- `sakila-data.sql` -- Populates the tables with sample data.
- `sakila.mwb` -- Workbench data model file for visual schema inspection.

Step 6: Install Sakila

Option A -- Using Workbench:

1. Open MySQL Workbench and connect to **Local MySQL 8.4**.
2. Go to **File -> Open SQL Script**.
3. Open `sakila-schema.sql` and **Execute** it.
4. Then open `sakila-data.sql` and **Execute** it.

Option B -- Using Terminal:

```
mysql -u root -p
```

Then run:

```
SOURCE /path/to/sakila-db/sakila-schema.sql;  
SOURCE /path/to/sakila-db/sakila-data.sql;
```

Always run the schema file first, then the data file.

Step 7: Verify Sakila Installation

```
SHOW DATABASES;  
-- You should see 'sakila' in the list  
  
USE sakila;  
  
SHOW TABLES;  
-- You should see all the Sakila tables  
  
SELECT * FROM actor LIMIT 10;  
  
SELECT COUNT(*) FROM film;  
-- Expected: 1000  
  
SELECT * FROM customer LIMIT 10;
```

Windows Setup

Step 1: Install MySQL Server 8.4 LTS

1. Go to: [MySQL Community Server Downloads](#)
2. Choose the Windows MSI package for MySQL 8.4.

3. Run the MSI installer.
4. During configuration, choose:

```
Configuration Type:  Development Computer
Port:                3306
Username:            root
Password:            <your chosen password>
```

5. Allow MySQL to run as a **Windows service**.

Step 2: Install MySQL Workbench

1. Go to: [MySQL Workbench Downloads](#)
2. Download and install the Windows package.
3. Connect using:

```
Hostname: localhost
Port:     3306
Username: root
```

Step 3: Install Sakila

Follow the same Sakila download and installation steps as described in the macOS section above (Steps 5-7).

Ubuntu / Debian Linux Setup

Use the official **MySQL APT Repository** rather than the distribution's default repository.

```
# After configuring the MySQL APT repository:
sudo apt-get update
sudo apt-get install mysql-server

# Check MySQL is running:
sudo systemctl status mysql

# Test connection:
mysql -u root -p

# Install Workbench:
sudo apt-get install mysql-workbench-community
```

Then install Sakila using the same `.sql` files (Steps 5-7 from the macOS section).

Fedora / RHEL / Rocky / AlmaLinux Setup

Use the official **MySQL Yum Repository**.

```
Add MySQL Yum Repository
|
Select MySQL 8.4 LTS
|
```

```
Install mysql-community-server
|
Start mysqld
|
Set/obtain temporary root password
|
Install Workbench
|
Install Sakila
```

Official installation documentation: [MySQL 8.4 Linux Installation Guide](#)

Sakila Works on All Operating Systems

The database files are OS-independent. You download `sakila-schema.sql`, `sakila-data.sql`, and `sakila.mwb`, and they work on any platform running MySQL Server.

```
macOS      --+
Windows    --+
Ubuntu     --+--> MySQL Server --> sakila
Fedora/RHEL --+
Any Linux  --+
```

Recommended Lab Environment

To minimize setup problems, the class is standardized on:

Platform	MySQL Server	MySQL Workbench	Database	Port
macOS	8.4 LTS	8.0.47	Sakila	3306
Windows	8.4 LTS	8.0.47	Sakila	3306
Linux	8.4 LTS	8.0.47	Sakila	3306

Installation Verification Checklist

Use this checklist to confirm your environment is ready:

- ☐ MySQL Community Server 8.4 LTS installed
- ☐ MySQL Workbench 8.0.47 installed
- ☐ Sakila Sample Database downloaded
- ☐ MySQL root password recorded safely
- ☐ Workbench connected to `localhost:3306`
- ☐ Sakila database installed (schema + data)
- ☐ `SELECT VERSION();` works
- ☐ `SHOW DATABASES;` shows `sakila`
- ☐ `USE sakila;` works
- ☐ `SELECT * FROM actor LIMIT 10;` returns results

Teaching Tip: Have every student first run `SELECT VERSION();` and `SHOW DATABASES;` , then `USE sakila;` `SHOW TABLES;` . This gives a quick "installation verification exercise" before beginning SQL classes.

If all items are checked, you are ready for the course.
